Vector Calculus and Complex Analysis Classwork (2025/5/8)

Name:

Student ID:

Consider $z \in C = \{2t + i(3 - t^2) : -1 \le t \le 1\}$. Show that $\left|\frac{1}{\ln z}\right|$ is max at $t = 1$. Find the upper bound of $\left|\int_C \frac{1}{\ln z} dz\right|$

Solution

$$\ln z = \ln|z| + iArg(z) + 2n\pi i$$

$$= \frac{1}{2}\ln|z|^2 + iArg(z) + 2n\pi i$$

$$= \frac{1}{2}\ln\left((2t)^2 + (3-t^2)^2\right) + iArg(z) + 2n\pi i$$

$$= \frac{1}{2}\ln\left(t^4 - 2t^2 + 9\right) + iArg(z) + 2n\pi i$$

$$= \frac{1}{2}\ln\left((1-t^2)^2 + 8\right) + iArg(z) + 2n\pi i$$

$$|\ln z| \geq \left|\frac{1}{2}\ln 8 + i\frac{\pi}{4}\right| = \frac{(\ln 8)^2}{4} + \frac{\pi^2}{16} \qquad (t=1, n=0)$$

$$\ell(C) = \int_{-1}^{1}\sqrt{x'(t)^2 + y'(t)^2}\,dt = \int_{-1}^{1}\sqrt{2^2 + (-2t)^2}\,dt = 2\int_{-1}^{1}\sqrt{1+t^2}\,dt$$

Let $I = \int_{-1}^{1}\sqrt{1+t^2}\,dt = \int_{-\frac{\pi}{4}}^{\frac{\pi}{4}}\sec\theta d\tan\theta$

$$\int_{-\frac{\pi}{4}}^{\frac{\pi}{4}}\sec\theta d\tan\theta = \left[\sec\theta\tan\theta\right]_{-\frac{\pi}{4}}^{\frac{\pi}{4}} - \int_{-\frac{\pi}{4}}^{\frac{\pi}{4}}\tan\theta d\sec\theta$$

$$\int_{-\frac{\pi}{4}}^{\frac{\pi}{4}}\sec^3\theta d\theta = 2\sqrt{2} - \int_{-\frac{\pi}{4}}^{\frac{\pi}{4}}\sec\theta\tan^2\theta d\theta = 2\sqrt{2} - \int_{-\frac{\pi}{4}}^{\frac{\pi}{4}}\sec\theta(\sec^2\theta - 1)d\theta$$

$$2\int_{-\frac{\pi}{4}}^{\frac{\pi}{4}}\sec^3\theta d\theta = 2\sqrt{2} + \int_{-\frac{\pi}{4}}^{\frac{\pi}{4}}\sec\theta d\theta = 2\sqrt{2} + \ln\left[\sec\theta + \tan\theta\right]_{-\frac{\pi}{4}}^{\frac{\pi}{4}} = 2\sqrt{2} + 2\ln\left[\sqrt{2}+1\right]$$

$$\ell(C) = 2I = 2\int_{-\frac{\pi}{4}}^{\frac{\pi}{4}}\sec\theta d\tan\theta = 2\int_{-\frac{\pi}{4}}^{\frac{\pi}{4}}\sec^3\theta d\theta = 2\sqrt{2} + 2\ln\left[\sqrt{2}+1\right]$$

$$\left|\int_C \frac{1}{\ln z}dz\right| \leq \frac{2\sqrt{2} + 2\ln\left[\sqrt{2}+1\right]}{\frac{(\ln 8)^2}{4} + \frac{\pi^2}{16}}$$